

AYMANE AIT DADS

AI Research Intern - Agentic AI | LLM Fine-Tuning · Supervised Fine-Tuning · Agentic Systems

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PROFESSIONAL SUMMARY

Achieved top 10% public leaderboard (0.80/1.0) in the NVIDIA Nemotron Reasoning Challenge (\$106K+ prize pool) by fine-tuning a 30B Mamba-Transformer MoE via LoRA ($r=32$, ~888M trainable parameters) with SFTTrainer and completion-only loss, directly demonstrating LLM fine-tuning and supervised fine-tuning expertise central to this role. Ranked 1st in the EURECOM class leaderboard for AI-generated image detection (0.791 AUC) using CLIP ViT-L/14 with dual learning-rate fine-tuning and submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark. Technical skills span expert Python, PyTorch, Hugging Face Transformers, vLLM, PEFT, prompt engineering, RAG, LLM orchestration, tool-calling, and multi-agent system design with Claude API and MCP server. Available for internship in Europe immediately; holds EU-compatible student status based in Sophia Antipolis, France.

CORE COMPETENCIES

LLM Fine-Tuning | Supervised Finetuning | Agentic AI Systems | Prompt Engineering | PyTorch & NLP | Hugging Face & vLLM | Multi-Agent Interactions | Ablation Studies

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Deployed a supervised fine-tuning pipeline combining Random Forest and K-Means on **100K+ fiber-optic network records** to classify network quality across thousands of nodes.
- Designed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output directly adopted by the network optimization team.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation using Python, Scikit-learn, and Pandas.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, bridging research findings with production decision-making.

PROJECTS

NVIDIA Nemotron Reasoning Challenge *Kaggle Competition · In Progress June 2026 · Top 10% Public Leaderboard · \$106K+ Prize Pool*

- Achieved **top 10% public leaderboard score of 0.80/1.0** (vs. 0.49 base, 0.86 best competitor) by fine-tuning a 30B Mamba-Transformer MoE model with LoRA ($r=32$, $\alpha=32$, ~888M trainable parameters) using SFTTrainer completion-only loss across 7,828 verified chain-of-thought examples for 6 logic puzzle types.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict dependency order - including monkey-patching the CUDA caching allocator, stubbing Mamba/Cutlass imports, and replacing Triton kernels with PyTorch fallbacks - and conducted 20+ single-variable ablations confirming that packing degraded score from 0.80 to 0.64 and a 12-layer LoRA hit a hard ceiling at 0.58.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Mamba · Triton

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · 1st in Class Leaderboard

- Ranked **1st in the EURECOM class private Kaggle leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates ($2e-6$ backbone, $5e-4$ head), and unfreezing up to 8 transformer blocks across 250K images from 25+ generator types.
- Submitted to the **NTIRE 2026 @ CVPR CodaBench international benchmark**, with ablation studies confirming 1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC) for out-of-distribution generalization, and a 3-model ensemble with 10-view TTA delivering the final submission score.

PyTorch · OpenCLIP · Hugging Face Transformers · FP16 Mixed Precision · Kaggle T4 GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Image Security, Cloud Computing, Statistics, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & Agentic AI: LoRA / QLoRA | SFTTrainer | vLLM | LLM Orchestration | RAG | Tool-Calling | MCP Server | LangChain

ML / DL Frameworks & NLP: PyTorch | Hugging Face Transformers | PEFT | TRL | Scikit-learn | OpenCLIP | Unsloth | Mamba-Transformer

Engineering & Data: Python (Expert) | CUDA Debugging | Triton | FP16/BF16 Mixed Precision | Git | Linux | Pandas | Power BI